

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: Database Management Systems
COURSE CODE: CSA05

COURSE OUTCOME COVERED

CO3: Analyze relational databases using functional dependencies, normalization (1NF to 5NF), and key constraints to identify redundancy and ensure data integrity. (BL2, BL3)

Activity Tool : MCQ Test (Low)
Assessment Tool Weightage: 30%

QUESTIONS		Total Marks: 50	
Q.No	Question	Bloom's Level	Marks
1	Which of the following is true about 1NF? (a) Allows multi-valued attributes (b) Eliminates partial dependencies (c) All attributes must be atomic (d) Eliminates transitive dependencies	BL2 – Understand	5
2	A relation is in 2NF if it is in 1NF and: (a) Has no transitive dependencies (b) Has no partial dependencies on primary key (c) Every attribute is a prime attribute (d) Has no multi-valued dependencies	BL2 – Understand	5
3	Which normal form deals with multi-valued dependencies? (a) 2NF (b) 3NF (c) BCNF (d) 4NF	BL2 – Understand	5
4	BCNF is a stricter version of: (a) 1NF (b) 2NF (c) 3NF (d) 4NF	BL2 – Understand	5
5	Functional dependency $X \rightarrow Y$ means: (a) Y determines X (b) X determines Y (c) X and Y are unrelated (d) Both determine each other	BL2 – Understand	5
6	Which of the following anomaly is NOT associated with unnormalized relations? (a) Insertion anomaly (b) Deletion anomaly (c) Update anomaly (d) Retrieval anomaly	BL2 – Understand	5
7	A superkey is: (a) A minimal set of attributes that uniquely identifies tuples (b) Any set of attributes that uniquely identifies tuples (c) An attribute that depends on a non-prime attribute (d) A foreign key reference	BL2 – Understand	5
8	If R(A,B,C) has FD: $A \rightarrow B$ and $B \rightarrow C$, then C is: (a) Partially dependent on A (b) Fully dependent on A (c) Transitively dependent on A (d) Independently determined	BL2 – Understand	5
9	5NF is also known as: (a) Domain-Key Normal Form (b) Boyce-Codd Normal Form (c) Project-Join Normal Form (d) Element Normal Form	BL2 – Understand	5
10	Lossless-join decomposition ensures: (a) No data is lost during decomposition (b) All FDs are preserved (c) The relation is in BCNF (d) No null values exist	BL2 – Understand	5

Code of Conduct:

I E.Jeffrey Samuels(192524412) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:Jeffrey Samuels.E

RUBRICS FOR CALCULATION:

MCQ					
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding; several incorrect responses	Lacks understanding ; majority of answers incorrect
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers
Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts
Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time
Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions

SIMATS ENGINEERING

ASSESSMENT TOOL 3 — CO3 MCQ TEST

Questions with Answers

Q1. Which of the following is true about 1NF?

- (a) Allows multi-valued attributes
- (b) Eliminates partial dependencies
- (c) All attributes must be atomic
- (d) Eliminates transitive dependencies

Ans: (c) All attributes must be atomic

Explanation: 1NF requires every attribute to hold a single, atomic (indivisible) value — no repeating groups or multi-valued attributes.

Q2. A relation is in 2NF if it is in 1NF and:

- (a) Has no transitive dependencies
- (b) Has no partial dependencies on primary key
- (c) Every attribute is a prime attribute
- (d) Has no multi-valued dependencies

Ans: (b) Has no partial dependencies on primary key

Explanation: $2NF = 1NF + \text{no partial dependency of any non-prime attribute on part of a composite candidate key}$.

Q3. Which normal form deals with multi-valued dependencies?

- (a) 2NF
- (b) 3NF
- (c) BCNF
- (d) 4NF

Ans: (d) 4NF

Explanation: 4NF removes non-trivial multi-valued dependencies (MVDs).

Q4. BCNF is a stricter version of:

- (a) 1NF
- (b) 2NF
- (c) 3NF
- (d) 4NF

Ans: (c) 3NF

Explanation: BCNF tightens 3NF by requiring every determinant to be a candidate key.

Q5. Functional dependency $X \rightarrow Y$ means:

- (a) Y determines X
- (b) X determines Y
- (c) X and Y are unrelated
- (d) Both determine each other

Ans: (b) X determines Y

Explanation: By definition, $X \rightarrow Y$ means the value of X determines (functionally decides) the value of Y.

Q6. Which of the following anomaly is NOT associated with unnormalized relations?

- (a) Insertion anomaly
- (b) Deletion anomaly
- (c) Update anomaly
- (d) Retrieval anomaly

Ans: (d) Retrieval anomaly

Explanation: The three classic anomalies from redundancy are insertion, deletion, and update anomalies. "Retrieval anomaly" is not a recognized category.

Q7. A superkey is:

- (a) A minimal set of attributes that uniquely identifies tuples
- (b) Any set of attributes that uniquely identifies tuples ✓ Correct Answer
- (c) An attribute that depends on a non-prime attribute
- (d) A foreign key reference

Ans: (b) Any set of attributes that uniquely identifies tuples

Explanation: A superkey only needs to guarantee uniqueness — minimality is what turns a superkey into a candidate key (option a describes a candidate key, not a superkey).

Q8. If $R(A,B,C)$ has FD: $A \rightarrow B$ and $B \rightarrow C$, then C is:

- (a) Partially dependent on A
- (b) Fully dependent on A
- (c) Transitively dependent on A
- (d) Independently determined

Ans: (c) Transitively dependent on A

Explanation: $A \rightarrow B$ and $B \rightarrow C$ together give $A \rightarrow C$ indirectly through B — the textbook definition of a transitive dependency.

Q9. 5NF is also known as:

- (a) Domain-Key Normal Form
- (b) Boyce-Codd Normal Form
- (c) Project-Join Normal Form
- (d) Element Normal Form

Ans: (c) Project-Join Normal Form

Explanation: 5NF is formally called Project-Join Normal Form (PJNF).

Q10. Lossless-join decomposition ensures:

- (a) No data is lost during decomposition
- (b) All FDs are preserved
- (c) The relation is in BCNF
- (d) No null values exist

Ans: a) No data is lost during decomposition

Explanation: Lossless-join is defined precisely as: joining the decomposed relations back together reproduces the original relation exactly, with no spurious or lost tuples.